Lab 3

Problem 1:

Write a C program that:

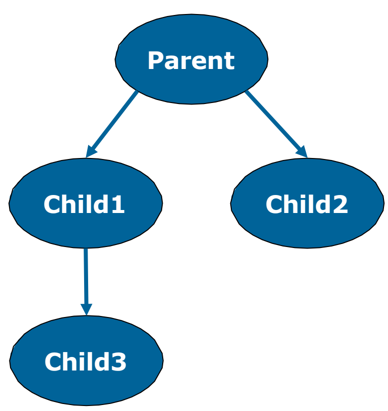
1. Displays the PID and the PPID of the main parent process
2. Forks a child
3. Checks if the fork system call failed, raise an error and exit the program
4. The child should first display its PID and its parent PPID
5. The parent should wait for the child to terminate. The parent should first display its PID, child PID and its parent PPID. The parent then should retrieve the exit status of its child and displays it. The parent should display a corresponding message such as ”Child Failed” with the child status, otherwise ”Child Succeeded” based on the exit status of the child.

Sample Output:

Main Process PID=1208, PPID=1114
Child Process PID= 1209, PPID=1208
Parent Process PID= 1208, child PID= 1209, PPID=1114
Child exit code=0
Child succeeded!

Problem 2:

Write a C program that will create the following processes and in every process will display the PID and the PPID.



Sample Output:

Child3 Process PID= 1359, PPID=1358
Child1 Process PID= 1358, PPID=1357
Child2 Process PID= 1360, PPID=1357
Parent Process PID= 1357, PPID=1114